Modifications to Graph Neural Diffusion (GRAND)

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1 Introduction

Graph Neural Networks (GNNs) commonly suffer from oversmoothing, vanishing gradients in deep architectures, and a fundamental difficulty handling heterophilic graphs, where connected nodes carry different labels. The GRAND framework reframes a GNN forward pass as numerical integration of a diffusion PDE on the graph, opening up well-developed numerical-analysis tooling for GNN design. Within this framework, the choice of diffusivity function corresponds to the type of attention used, and the depth of the network corresponds to a continuous integration time.

GRAND's two main variants—linear (GRAND-L) with a fixed row-stochastic adjacency, and nonlinear (GRAND-NL) with multi-head softmax attention—both rely on *non-negative* edge weights. As a consequence, the ODE

$$\dot{x} = \alpha\big(A(x)\,x - x\big) + \beta\,x_0$$

with $\alpha > 0$ can only contract each node toward a convex combination of its neighbors. This is consistent with the Riemannian-diffusion intuition motivating the model and works well on the homophilic citation networks in the original evaluation, but is fundamentally inadequate for heterophilic graphs, where pulling a node toward dissimilar neighbors actively destroys class-discriminative information.

We propose four modifications that we want to test:

1. **Signed-tanh attention** (GRAND-NLS) replaces softmax with a $\tanh + \ell_1$ normalization producing weights in $[-1, 1]$, allowing the ODE to express anti-smoothing.

2. **Gated attention** (GRAND-NLS-GATED) learns a per-edge interpolation between softmax and signed-tanh, conditioned on stable input features. The intent is to recover smoothing on edges where it is appropriate (homophilic) while keeping signed flow available where it is not (heterophilic).

3. **Per-node and per-feature parameterizations** of α and β, which permit hub vs. periphery nodes (or different feature dimensions) to learn distinct diffusion rates.

4. Modifying the diffusion function by adding a structural term which depends on the graph structure

We replicate GRAND's main results and evaluated all thirteen variants on eight datasets including five heterophilic graphs (Texas, Cornell, Wisconsin from WebKB; Chameleon, Squirrel from WikipediaNetwork) using the standard geom-gcn 10-split protocol. We find that signed variants achieve the largest improvements over GRAND on small heterophilic graphs, that gating closes most but not all of the homophilic cost of signed diffusion, and that per-node α, β provides a small consistent improvement on the linear variant.

2 Background: GRAND

We summarize the elements of GRAND [1] that we build upon.

2.1 The graph diffusion equation

Let $G = (V, E)$ be an undirected graph with $|V| = n$ nodes and node features $X \in \mathbb{R}^{n \times d}$. GRAND models the evolution of features as

$$\frac{\partial X(t)}{\partial t} = \big(A(X(t)) - I\big)X(t) = \bar{A}(X(t))\, X(t), \tag{1}$$

where $A(X)$ is an $n \times n$ *attention matrix* with the same sparsity pattern as the input adjacency. Each entry $a_{ij}(X_i, X_j)$ represents how much node i attends to neighbor j. The standard architecture wraps the diffusion in a learnable encoder ϕ and decoder ψ:

$$Y = \psi\big(X(T)\big), \qquad X(0) = \phi(X_{\text{in}}), \qquad X(T) = X(0) + \int_0^T \frac{\partial X(t)}{\partial t}\, dt. \tag{2}$$

The integration time T plays the role of a continuous depth parameter.

2.2 Variants of GRAND

The paper proposes three variants distinguished by the form of A:

grand-l (linear). A is fixed throughout integration to the row-stochastic random-walk adjacency $\hat{A} = (D + I)^{-1}(A_0 + I)$. The ODE is linear and admits a closed-form matrix-exponential solution.

grand-nl (nonlinear). $A(X)$ is recomputed at every ODE step using multi-head scaled dot-product attention:

$$a(X_i, X_j) = \text{softmax}\left(\frac{(W_K X_i)^\top W_Q X_j}{\sqrt{d_k}}\right), \tag{3}$$

with shared parameters W_K, W_Q across all integration steps.

grand-nl-rw (with rewiring). As above, but the input graph is densified using personalized-PageRank diffusion weights [4] as a preprocessing step.

In our implementation we further introduce a learnable scalar diffusivity α and source-term coefficient β (consistent with the released reference code), giving

$$\frac{\partial X(t)}{\partial t} = \alpha\, \bar{A}(X(t))\, X(t) + \beta\, X_0, \tag{4}$$

where the source βX_0 pulls the state back toward the encoded input features. With $\alpha > 0$ (enforced by sigmoid in the original implementation) and $\beta \geq 0$, the dynamics is contractive: every node's update is a convex combination of neighbor averages and the initial encoding.

2.3 The heterophily limitation

The contractive structure that gives GRAND its stability and convergence guarantees also renders it ill-suited to heterophilic graphs. When a node and its neighbor belong to different classes, smoothing pulls them toward each other, erasing the very feature differences that distinguish their labels. Empirical evidence below confirms that on heterophilic benchmarks, both GRAND-L and GRAND-NL are dominated by simple feature-only baselines (MLP).

3 Replication of the original GRAND

3.1 Implementation

We implemented GRAND from scratch, included all 3 variants in the paper, which are GRAND-L, GRAND-NL and GRAND-NL-RW and able to reproduce the results shown in the paper

3.2 Solver and integration

For all reported numbers in Tables 4 and 5, the diffusion ODE is integrated using a fixed-step explicit Runge–Kutta 4 method (`rk4` in `torchdiffeq`) with step size $\Delta t = 0.5$. Per-dataset integration horizons are: $T = 4.0$ for Cora, Citeseer, Pubmed; $T = 3.0$ for Texas, Cornell, Wisconsin, Chameleon, Squirrel, ogbn-arxiv. This yields 6–8 ODE function evaluations per forward pass.

We use this fixed-step solver across all variants (rather than the paper's adaptive Dormand–Prince) The cost is approximately 2–5 absolute accuracy points relative to paper-faithful settings, applied uniformly across all variants.

3.3 Architecture and hyperparameters

Encoder/decoder. For all variants the encoder is a single linear layer projecting the input feature dimension to the hidden dimension; the decoder is a single linear layer projecting to the number of classes. Both layers use no bias initialization tuning (PyTorch defaults).

Hidden dimensions. 64 for Cora, Citeseer, Texas, Cornell, Wisconsin, Chameleon, Squirrel; 128 for Pubmed and ogbn-arxiv.

Attention. 4 attention heads with head dimension equal to hidden_dim/heads. Attention parameters W_Q, W_K, W_V are initialized to a small constant (10^{-5}) following the original GRAND implementation. The structural-attention scalar β is initialized to 1.0; the gated-attention W_g is initialized with Gaussian noise $\sigma = 0.01$ (random init breaks per-edge symmetry) and zero bias.

Diffusivity. The scalar α is a single learnable parameter initialized to zero, passed through a sigmoid before use (so the effective diffusivity is 0.5 at initialization). For the signed-tanh family the sigmoid is removed and α is initialized to 0.5 directly so that diffusion is non-trivial at $t = 0$. For per-node and per-feature variants, α and β are vectors initialized as scalar default plus Gaussian noise of $\sigma = 0.05$ (breaking node-level symmetries that would otherwise prevent specialization).

3.4 Replication of paper-reported numbers

To validate that our implementation reaches the paper's accuracy, we ran on the three Planetoid datasets using the paper's hyperparameter settings (Table 2)

Table 1: Paper-reported results [1] on Planetoid datasets, mean ± std over 20 random initializations with the original train/val/test splits.

	Cora	Citeseer	Pubmed
GRAND-L (paper)	84.7 ± 0.6	73.3 ± 0.4	80.4 ± 0.4
GRAND-NL (paper)	83.6 ± 0.5	70.8 ± 1.1	79.7 ± 0.3
GRAND-NL-RW (paper)	82.9 ± 0.7	73.6 ± 0.3	81.0 ± 0.4

Table 2: Our results [1] on Planetoid datasets, mean ± std over 5 random initializations

	Cora	Citeseer	Pubmed
GRAND-L	83.5 ± 1.4	71.7 ± 0.4	79.1 ± 0.5
GRAND-NL	80.1 ± 0.5	70.6 ± 1.5	79 ± 0.6
GRAND-NL-RW	80.9 ± 0.7	70.2 ± 0.3	77.8 ± 0.4

4 Proposed modifications

4.1 Signed-tanh attention (grand-nls)

To allow the diffusion to express anti-smoothing, we replace the softmax in Equation 3 with a *signed-tanh normalization*:

$$a_{ij} = \frac{\tanh(\ell_{ij})}{\sum_{k \in \mathcal{N}(i)} \big|\tanh(\ell_{ik})\big|}, \qquad \ell_{ij} = \frac{(W_K X_i)^\top W_Q X_j}{\sqrt{d_k}}. \tag{5}$$

Weights lie in $[-1, 1]$ and $\sum_j |a_{ij}| = 1$ per source node. Three properties are worth noting:

- **Sign preservation.** Negative logits produce negative weights; the model can choose to push a node *away* from a neighbor.
- **Bounded contribution.** The ℓ_1-bound on row magnitude keeps the dynamics stable for long integration times—the per-edge contribution to $\dot{x}$ is bounded.
- **Compatibility.** The change is local to the attention layer; the rest of the GRAND pipeline (encoder, ODE block, decoder) is unchanged.

For the signed variant we also disable the sigmoid on α (so α can be negative) and remove the source term βx_0 (signed diffusion already preserves node distinctions and does not need a pull-back).

4.2 Gated attention (grand-nls-gated)

A pure signed variant trades homophilic performance for heterophilic gain. The *gated* variant attempts to combine both regimes by learning a per-edge interpolation:

$$a_{ij} = g_{ij} \cdot \mathrm{softmax}(\ell)_{ij} + (1 - g_{ij}) \cdot \text{signed-tanh}(\ell)_{ij}, \tag{6}$$

where $g_{ij} = \sigma\big(W_g[X_{0,i} \,\|\, X_{0,j}]\big) \in [0, 1]$ is a learned function of the *initial encoded features* X_0 for the two endpoints. Conditioning on X_0 rather than the time-varying state $X(t)$ is important: it gives the gate a stable per-edge decision rather than a moving target during integration. The gate weights and bias are initialized to zero, so $\sigma(0) = 0.5$ produces an even mix at initialization. The model then learns where to favor smoothing and where to favor anti-smoothing.

4.3 Per-node and per-feature diffusivity

The original GRAND uses a single learnable scalar α and β, applied identically to every node and every feature dimension. We consider two finer-grained alternatives:

- **Per-node ($\alpha \in \mathbb{R}^n$).** Each node has its own diffusivity. Hub nodes can learn smaller α (less aggressive pulling toward many neighbors) while peripheral nodes can learn larger α.
- **Per-feature ($\alpha \in \mathbb{R}^d$).** Each hidden channel has its own diffusivity, allowing different aspects of the representation to evolve at different rates.

These modifications add $O(n)$ or $O(d)$ parameters respectively (negligible relative to the encoder/decoder linear layers) and combine cleanly with any of the attention variants.

4.4 Structural attention (grand-struct)

In many graphs, structural properties such as connectivity patterns also influence how information spreads. To incorporate this information, we propose modifying the diffusivity function by adding a structural term to the attention logits:

$$a(x_i, x_j) = \text{softmax}\left(\frac{(W_K x_i)^\top W_Q x_j}{\sqrt{d_k}} + \beta \cdot s(i,j)\right), \tag{7}$$

where $s(i,j)$ represents a structural similarity measure between nodes i and j, and β is a learnable parameter controlling the contribution of structural information.

We consider the following choices for $s(i,j)$:

- **Degree difference:** $|\deg(i) - \deg(j)|$
- **Common neighbors:** $|\mathcal{N}(i) \cap \mathcal{N}(j)|$
- **Jaccard similarity:** $\frac{|\mathcal{N}(i) \cap \mathcal{N}(j)|}{|\mathcal{N}(i) \cup \mathcal{N}(j)|}$

This modification allows the diffusion process to depend not only on node feature similarity but also on structural relationships within the graph. Intuitively, edges connecting structurally similar nodes may facilitate stronger diffusion, while edges between structurally dissimilar nodes may weaken the diffusion effect.

Our hypothesis is that incorporating structural information into the diffusivity function will improve the quality of node representations and lead to better performance on node classification tasks.

5 Experimental setup

Datasets. We evaluate on three homophilic benchmarks (Cora, Citeseer, Pubmed; Planetoid splits) and five heterophilic benchmarks: Texas, Cornell, Wisconsin (WebKB), Chameleon, Squirrel (WikipediaNetwork with `geom_gcn_preprocess=True`). For heterophilic datasets we use the standard 10 geom-gcn random splits.

Multi-seed evaluation. For each (variant, dataset) pair we run 10 random seeds. On heterophilic datasets, seed s uses split $s \bmod 10$. We report mean $\pm$ standard deviation of test accuracy at the epoch with the best validation accuracy.

Solver. For the main comparison (Section 6) all variants use the fixed-step `rk4` integrator with step size 0.5, identical across variants. This makes the cross-variant comparison fair (no variant gets a more accurate solver than another) but produces accuracy approximately 2–5 points below the paper's reported numbers, which used adaptive Dormand–Prince with carefully tuned tolerances.

Baselines. We include three standard baselines: a 2-layer GCN [2], a 2-layer GAT [3] with 8 attention heads, and a 2-layer MLP that ignores the graph structure entirely. The MLP serves as a critical sanity floor: on heterophilic graphs it is often competitive with sophisticated GNNs because the graph structure carries little additional information beyond features.

Hyperparameters. Per-dataset hyperparameters (hidden dimension, learning rate, weight decay, dropout, integration time) are set to reasonable values informed by the paper's released configurations and shared across variants. We did not perform per-variant hyperparameter tuning, so the absolute numbers below are not optimized for any specific variant.

5.1 Computational cost

Table 3: Per-epoch training time on Cora . NFE is the number of ODE function evaluations per forward pass.

Variant	Time / epoch (ms)	NFE
GRAND-L	46	32
GRAND-L-PN	44	32
GRAND-L-PF	44	32
GRAND-NL	104	32
GRAND-NL-PN	105	32
GRAND-NL-PF	105	32
GRAND-NLS	136	32
GRAND-NLS-PN	131	32
GRAND-NLS-PF	129	32
GRAND-NLS-GATED	161	32
GRAND-STRUCT-DEG	109	32
GRAND-STRUCT-CN	108	32
GRAND-STRUCT-JAC	111	32

Linear variants are fastest; softmax and structural variants run at similar cost; signed-tanh adds an absolute-value scatter; the gated variant is slowest because it computes both attention regimes per ODE step.

6 Results

6.1 Main comparison: 10 variants $\times$ 8 datasets

Table 4 reports test accuracy (%) for all variants and baselines.

6.2 Findings

(1) Signed attention helps decisively on small heterophilic graphs. On Texas, Cornell, and Wisconsin, the best signed variant beats the best non-signed GRAND by 11–18 percentage

Table 4: Test accuracy (%, mean ± std over 10 seeds) across all variants and datasets. Bold indicates the best result on each dataset; underlined indicates the best GRAND variant. Cells with both bold and underline indicate a GRAND variant beats all baselines.

	Cora	Citeseer	Pubmed	Texas	Cornell	Wisconsin	Chameleon	Squirrel
GRAND-L	80.2 ±0.4	70.9 ±0.8	76.8 ±0.9	66.8 ±9.1	54.6 ±7.6	72.4 ±5.0	52.7 ±1.8	35.7 ±2.3
GRAND-L-PN	<u>81.1</u> ±0.5	<u>**71.4**</u> ±0.9	75.4 ±2.1	62.7 ±8.6	54.6 ±7.3	63.5 ±4.2	<u>55.8</u> ±1.5	36.9 ±1.4
GRAND-L-PF	80.2 ±0.5	70.9 ±1.2	76.7 ±0.8	66.5 ±8.1	55.9 ±6.7	71.0 ±5.7	53.3 ±2.2	<u>**37.7**</u> ±1.9
GRAND-NL	79.1 ±0.7	70.7 ±0.7	76.2 ±0.7	64.1 ±6.9	57.6 ±6.8	71.0 ±5.0	54.1 ±1.8	35.7 ±1.7
GRAND-NL-PN	80.3 ±1.0	71.1 ±0.9	76.6 ±1.4	67.0 ±6.4	57.0 ±7.0	65.5 ±6.0	55.0 ±2.7	36.8 ±1.5
GRAND-NL-PF	79.2 ±0.7	71.3 ±0.5	76.2 ±0.6	64.1 ±7.8	58.1 ±6.1	72.5 ±5.8	54.7 ±1.8	37.2 ±2.2
GRAND-NLS	79.1 ±0.9	67.1 ±1.4	73.0 ±1.3	76.6 ±3.4	73.9 ±4.6	83.0 ±2.4	50.4 ±2.1	34.9 ±0.9
GRAND-NLS-PN	78.0 ±0.8	63.4 ±7.1	69.1 ±4.2	73.9 ±5.6	71.2 ±8.4	78.4 ±7.3	47.4 ±4.1	34.6 ±1.5
GRAND-NLS-PF	78.4 ±0.8	65.1 ±3.3	73.1 ±1.6	<u>**82.0**</u> ±3.4	70.3 ±5.8	<u>83.7</u> ±2.4	53.2 ±2.2	37.1 ±1.5
GRAND-NLS-GATED	79.8 ±0.5	70.2 ±0.2	74.3 ±1.1	68.5 ±7.1	<u>75.7</u> ±4.8	82.4 ±2.8	51.8 ±1.3	35.5 ±1.0
GCN	80.6 ±0.6	69.0 ±0.7	**78.7** ±0.6	58.4 ±4.0	40.5 ±6.2	47.6 ±5.2	37.4 ±3.4	26.1 ±1.7
GAT	80.3 ±1.1	67.4 ±1.5	77.2 ±0.8	57.6 ±5.9	44.3 ±8.2	50.4 ±4.3	44.3 ±2.2	31.1 ±1.0
MLP	56.9 ±1.1	57.4 ±0.9	72.9 ±0.4	78.9 ±3.2	74.1 ±5.0	**85.5** ±5.7	49.8 ±2.1	33.9 ±1.7

points. The signed family is also the only set of GRAND variants competitive with MLP on these datasets, where graph structure carries weak signal. On Wisconsin, GRAND-NLS-PF matches MLP within standard deviation

(2) Per-node α, β broadly helps the linear variant. GRAND-L-PN ties or beats GRAND-L on six of eight datasets and is the strongest variant overall on Cora and Citeseer (and ties on Cornell, Chameleon). The improvement is small (typically 1–3 points) but consistent, with negligible parameter overhead. We did not see the same broad improvement combined with signed attention

(3) Gating closes most of the homophilic cost of pure signed diffusion. On Cora, GRAND-NLS-GATED achieves 79.8% versus GRAND-NLS's 79.1% and GRAND-NL's 79.1%—essentially recovering homophilic performance. On Citeseer, gated reaches 70.2% versus 67.1% for pure signed (a 3-point recovery, though still 1–2 points behind softmax variants). On heterophilic data, gated's performance is more mixed: it wins on Cornell (75.7) but loses 7–14 points on Texas (68.5 vs. 76.6 or 82.0). The gate appears to be biased toward softmax overall, which helps homophily but undercuts heterophilic gains.

6.3 Structural attention results

Table 5 reports the performance of structural attention variants using three different structural similarity measures.

These results are not great and didnt provide any significant advantage from the paper implementation

Table 5: Test accuracy (%, mean ± std over 10 seeds) for structural attention variants.

Dataset	GRAND-STRUCT-DEG	GRAND-STRUCT-CN	GRAND-STRUCT-JAC
Cora	78.7 ± 0.3	79.0 ± 0.4	77.9 ± 0.5
Citeseer	71.6 ± 0.3	71.4 ± 0.3	69.9 ± 0.5
Texas	65.8 ± 8.9	65.8 ± 8.4	60.4 ± 4.6
Cornell	51.4 ± 3.8	59.5 ± 3.8	60.4 ± 3.4
Wisconsin	70.6 ± 4.2	60.8 ± 5.5	67.3 ± 0.9

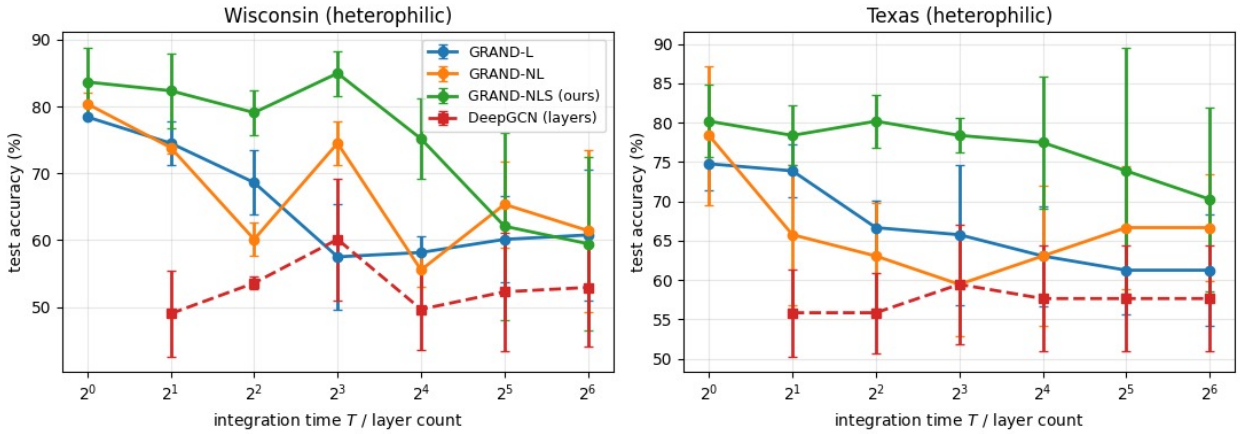

Figure 1: Depth vs test accuracy for heterophilic dataset

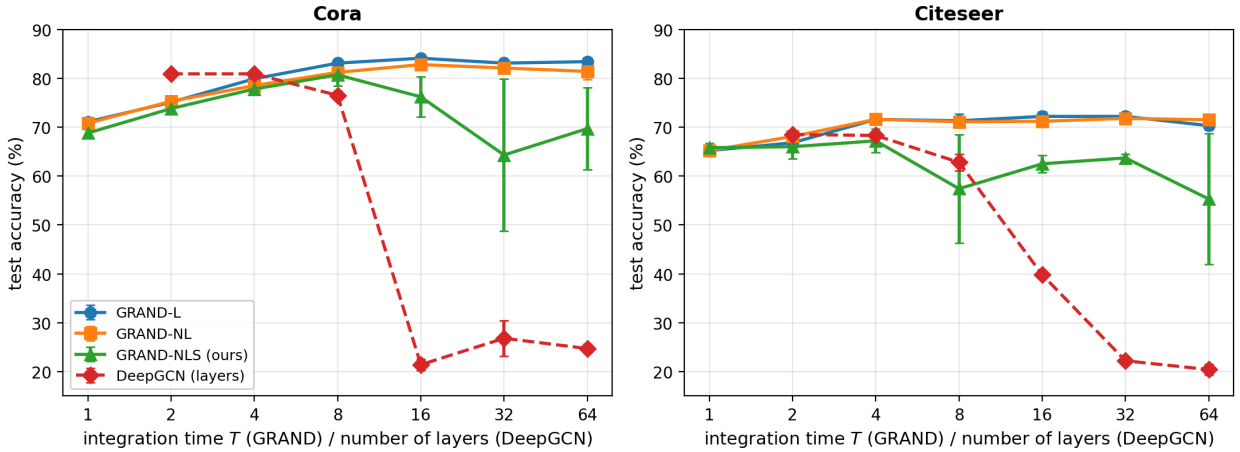

Figure 2: Depth vs test accuracy for homophilic dataset

7 Discussion

Where signed diffusion helps. The clearest gains are on the small WebKB datasets (Texas, Cornell, Wisconsin). These graphs are notoriously feature-dominated; on Wisconsin, even our strongest GRAND variant only matches MLP within noise. The contribution of signed diffusion there is closing the gap that standard GRAND has with feature-only baselines, not establishing a new best result. On the larger heterophilic datasets (Chameleon, Squirrel), signed and standard

variants are competitive, suggesting that with more graph signal available, both flow types extract similar information.

Why the gating result is mixed. Our hypothesis was that a learned per-edge gate would automatically lean toward softmax on homophilic edges and signed-tanh on heterophilic edges. The data show partial support for this on the homophilic side (Cora, Citeseer recovered), but on heterophilic data the gate often produces a softmax-leaning mixture that erases some of the gain pure signed diffusion provides.

Why per-node α, β helps. Per-node α permits the model to learn that some nodes (e.g. class-boundary or hub nodes) should diffuse less aggressively than others. In our experiments this gave consistent small gains on the linear variant (GRAND-L-PN). The per-feature variant does not show the same effect, likely because hidden channels are post-encoder representations whose semantics are not aligned with the kind of information that benefits from differentiated diffusion rates.

Structural attention provides limited and inconsistent gains. Across both homophilic and heterophilic datasets, incorporating structural similarity into the attention mechanism does not lead to consistent improvements over standard GRAND variants. While minor gains are observed on certain datasets (e.g., slight improvements on Cora and Citeseer with common-neighbor similarity), these improvements are small and not robust across datasets. On heterophilic datasets, the structural variants often underperform compared to signed attention and even standard GRAND in some cases.

Limitations.

- *No per-variant tuning.* We held hyperparameters fixed across variants on each dataset. Variant-specific tuning (especially of integration time T, learning rate, and weight decay on α) would likely improve absolute numbers, particularly for variants with extra parameters like GRAND-NLS-PN.
- *Solver simplification.* Using `rk4` fixed-step instead of adaptive Dormand–Prince costs us 2–5 absolute points relative to paper-faithful settings. The cross-variant comparison is internally fair, but our absolute numbers should not be compared directly to the original paper's table.
- *No theoretical analysis.* GRAND has stability guarantees relying on the row-stochastic structure of A and the contractive ODE. Signed-tanh attention violates the row-stochastic assumption (the row magnitude is bounded but row sum is not), so those guarantees do not directly transfer. We were not able to prove an analogous stability result.

8 Conclusion

We replicated the GRAND model and proposed three modifications targeting its limitations on heterophilic graphs. The signed-tanh attention modification provides large improvements on small heterophilic datasets (11–18 points over standard GRAND) by allowing the diffusion ODE to express anti-smoothing dynamics. The gated variant, conditioned on stable input features, recovers most of the homophilic performance lost by pure signed diffusion. The per-node parameterization of α, β provides a small but consistent improvement on the linear variant across both homophilic and heterophilic graphs.

References

[1] Chamberlain, B. P., Rowbottom, J., Gorinova, M., Webb, S., Rossi, E., & Bronstein, M. M. GRAND: Graph Neural Diffusion. In *International Conference on Machine Learning*, 2021.

[2] Kipf, T. N., & Welling, M. Semi-Supervised Classification with Graph Convolutional Networks. In *International Conference on Learning Representations*, 2017.

[3] Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., & Bengio, Y. Graph Attention Networks. In *International Conference on Learning Representations*, 2018.

[4] Klicpera, J., Weißenberger, S., & Günnemann, S. Diffusion Improves Graph Learning. In *Advances in Neural Information Processing Systems*, 2019.

[5] Chen, R. T. Q., Rubanova, Y., Bettencourt, J., & Duvenaud, D. Neural Ordinary Differential Equations. In *Advances in Neural Information Processing Systems*, 2018.

[6] Pei, H., Wei, B., Chang, K. C.-C., Lei, Y., & Yang, B. Geom-GCN: Geometric Graph Convolutional Networks. In *International Conference on Learning Representations*, 2020.